



Capstone Project 1
CMU-SE 450

Architecture Design
Version 1.2
Date: 13/12/2021

Online Clinic Booking System

Submitted by

**Truong Gia Huy
Ngo Van Kha
Le Van Xuan
Tran Nguyen Thanh Hong**

**Capstone Project 1 - Mentor
Master Nguyen Tan Thuan
Signature:**

Date:

A handwritten signature in black ink, appearing to be 'Thuan', written over a horizontal line.

Project Information

Project acronym	OnClinic		
Project Title	Online Clinic Booking System		
Start Date	23 Aug 2021	End Date	12 Dec 2021
Lead Institution	International School, Duy Tan University		
Project Mentor	Master Thuan, Nguyen Tan Email: thuannt.it.dtu@gmail.com		
Scrum master / Project Leader & contact details	Ngo Van Kha – 24211210573 Email: ngokha437@gmail.com Tel: 0935950384		
Partner Organization	N/A		
Team Members			
Student ID	Name	Email	Tel
24211206450	Truong Gia Huy	truonggiahuy7799@gmail.com	0972173687
24211207159	Le Van Xuan	levanxuan5518@gmail.com	0905596460
24207106137	Tran Nguyen Thanh Hong	thanhhong24102000@gmail.com	0782363961

Document Information

Document Title	Architecture Design
Reporting Period	
Author(s)	Ngo Van Kha, Le Van Xuan, Truong Gia Huy
Team members	Name:
	Truong Gia Huy
	Le Van Xuan
	Ngo Van Kha
	Tran Nguyen Thanh Hong
Date	13/12/2021
File name	C1SE.37-ArchitectureDesign-OnClinic
Access	Project and CMU Program

Document History		
Version	Date	Comments
V1.0	04/09/2021	Create Architecture Document
V1.1	15/11/2021	Additional documents
V1.2	13/12/2021	Completing the document

Table of contents

Project Information	2
Document Information	3
Table of contents	4
1. Introduction	5
1.1. Project overview	5
1.2. Purpose	5
1.3. Business driver	5
2. Architecture driver	6
2.1. Business constraints.....	6
2.2. Technical constraints	6
2.3. Functional requirement	6
2.4. Quality attributes	7
2.4.1. Security	7
2.4.2. Reliability	7
2.4.3. Performance	8
2.4.4. Usability	9
2.4.5. Correctness	10
2.4.6. Modifiability	10
3. Architecture overview	11
3.1. Component and connector	11
3.2. Module view	13
3.3. Allocation view	15

1. Introduction

1.1. Project overview

The online clinic booking system will build an Android application to help people intelligently book an appointment. Online medical examination support application full of basic features to make healthcare more convenient than ever.

1.2. Purpose

This specification covers the following:

- Brief specification of the project, high-level requirement.
- Detail quality attribution.
- System context, sequence diagrams.
- The architecture is presented by various view types: Component and Connect, Module view, and Allocation view.

1.3. Business driver

Business Problems:

- Patients need a more convenient way to take care of their health amid the covid 19 pandemic and the 4.0 technology era.
- Patients need a reliable and intelligent tool to help choose the best clinic for them.
- Doctors need a system that can promote their clinic in the information technology era, where everyone owns a smartphone.

Business Need:

- Doctors can post their clinic on the system.
- The system assists patients to book the best clinic, the system will give them top suggestions.
- Doctors can examine and communicate with patients online through video calls, online prescriptions, online payments, clinic reviews.
- Administrators can manage users, clinics...

2. Architecture driver

2.1. Business constraints

- Sources: 4 people.
- The project was started on: 23/08/2021
- The project will be ended on: 12/12/2021
- The project will be finished in 112 days (301 hours).
- Cost: 4480\$

2.2. Technical constraints

Technical to develop:

- Programming Language: Java (use Android Studio), Python.
- Frameworks / Libraries: OpenCV, Real-Time Heart Rate Monitoring From Facial RGB Color Video Using Webcam by H. Rahman, M.U. Ahmed, S. Begum, P. Funk, Jitsi meet.
- Database Real-time Management System: Firebase

Environment:

- App environment: Android
- Developing environment: Android studio

2.3. Functional requirement

References to Product Backlog specification (Product Backlog document)

2.4. Quality attributes

2.4.1. Security

Scenario: When users (doctor or patient) forget their password and want to reset their password, they must provide the email registered in the system, within 1 minute, a confirmation email will be sent to the email address entered to reset the password forgotten	
Type	Security
Stimulus	provide the email registered in the system
Source of stimulus	Users
Environment	In runtime
Artifact stimulated	Application
Response	a confirmation email will be sent to the email address entered to reset the password forgotten
Response measure	within 1 minute

2.4.2. Reliability

Scenario: Users (doctor or patient) want to use the system efficiently, minimize the impact of errors, feel comfortable with the system, so the system will always immediately provide the best user experience	
Type	Reliability
Stimulus	want to use the system efficiently, minimize the impact of errors, feel comfortable with the system
Source of stimulus	Users
Environment	In runtime
Artifact stimulated	The system
Response	will provide the best user experience
Response measure	always immediately

2.4.3. Performance

Scenario: Immediately after users (doctor or patient) confirm the clinic booking, the system will confirm and send a notification to the doctor within 5 seconds	
Type	Performance
Stimulus	confirm the clinic booking
Source of stimulus	Users
Environment	In runtime
Artifact stimulated	The system
Response	will confirm and send a notification to the doctor
Response measure	within 5 seconds

Scenario: Users (doctor or patient) want to use the application smoothly. So that, the system must handle incoming events with a near-instant delay	
Type	Performance
Stimulus	want to use the application smoothly
Source of stimulus	Users
Environment	In runtime
Artifact stimulated	The system
Response	must handle incoming events
Response measure	a near-instant delay

2.4.4. Usability

Scenario: Patients enter their symptoms. The system will predict the disease and make a list of the most suitable clinics for them conveniently and quickly	
Type	Usability
Stimulus	enter their symptoms
Source of stimulus	Patients
Environment	In runtime
Artifact stimulated	The system
Response	will predict the disease and make a list of the most suitable clinics for them
Response measure	conveniently and quickly

Scenario: When a patient makes a booking, they can freely choose to see the doctor online or offline, the system will support booking on both forms.	
Type	Usability
Stimulus	makes a booking
Source of stimulus	A patient
Environment	In runtime
Artifact stimulated	Mobile devices
Response	will support booking
Response measure	on both forms

2.4.5. Correctness

Scenario: When the patient enters the symptom, the system will diagnose the patient's disease with a high percentage of accuracy	
Type	Correctness
Stimulus	enters the symptom
Source of stimulus	the patient
Environment	In runtime
Artifact stimulated	The system
Response	will diagnose the patient's disease
Response measure	a high percentage of accuracy

2.4.6. Modifiability

Scenario: The product manager wants to develop additional features and build a health social network in the next version. The system allows for expansion within 4 months of effort without affecting existing functions.	
Type	Modifiability
Stimulus	wants to develop additional features and build a healthy social network
Source of stimulus	The product manager
Environment	the next version
Artifact stimulated	The system
Response	Allows for an expansion
Response measure	4 person-months of effort without affecting existing functions

3. Architecture overview

This section shows the diagrams which bound our target system and describes the architecture and interaction between components

3.1. Component and connector

We mainly used a C&C view to argue and reason about architectural properties, quality attribute requirements, and functional requirements that the system must add here.

This view type partitions the system into components that have some runtime presence such as processes, objects, data stores, and connectors or that represent pathways of communication such as data flow and access to shared storage.

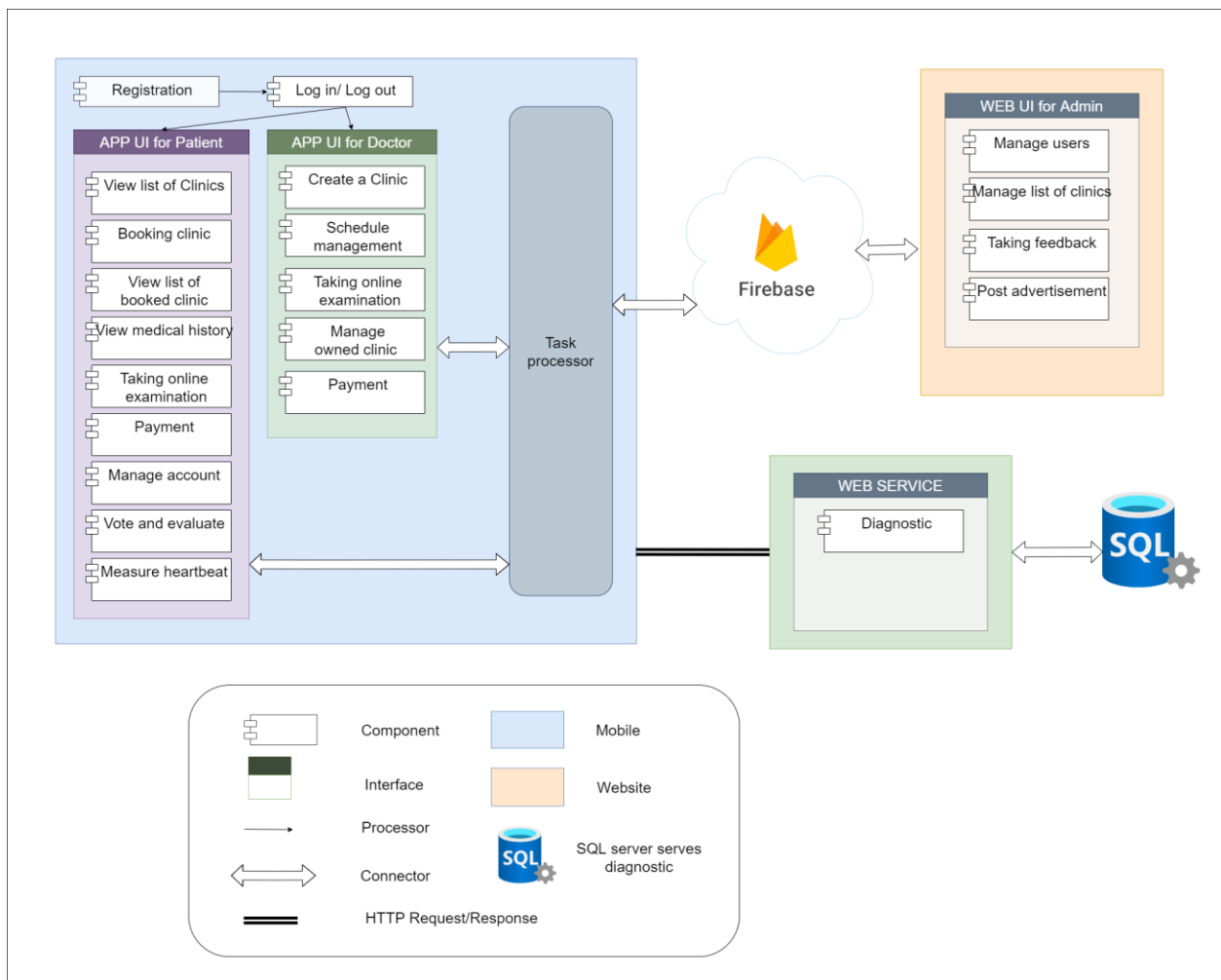


Figure 1: Component and connector view

Explication

Element	Responsibilities
App UI for Patient	Interface with patient-specific features
App UI for Doctor	Interface with doctor-specific features
DB connector	Contains code to communicate with firebase
Firebase service	Service providing real-time data storage solutions
Web UI for Admin	Manage users and clinics through Firebase

Description

After registration, the user will be assigned the default role of a patient, if desired, the patient can register as a partner with OnClinic, after being approved the account will switch to the role of a doctor.

After logging in, depending on the role of the account, the user will access the corresponding interface, each interface will have specific features specific to each role. Requests will be handled through the DB connector to connect and retrieve data with Firebase.

3.2. Module view

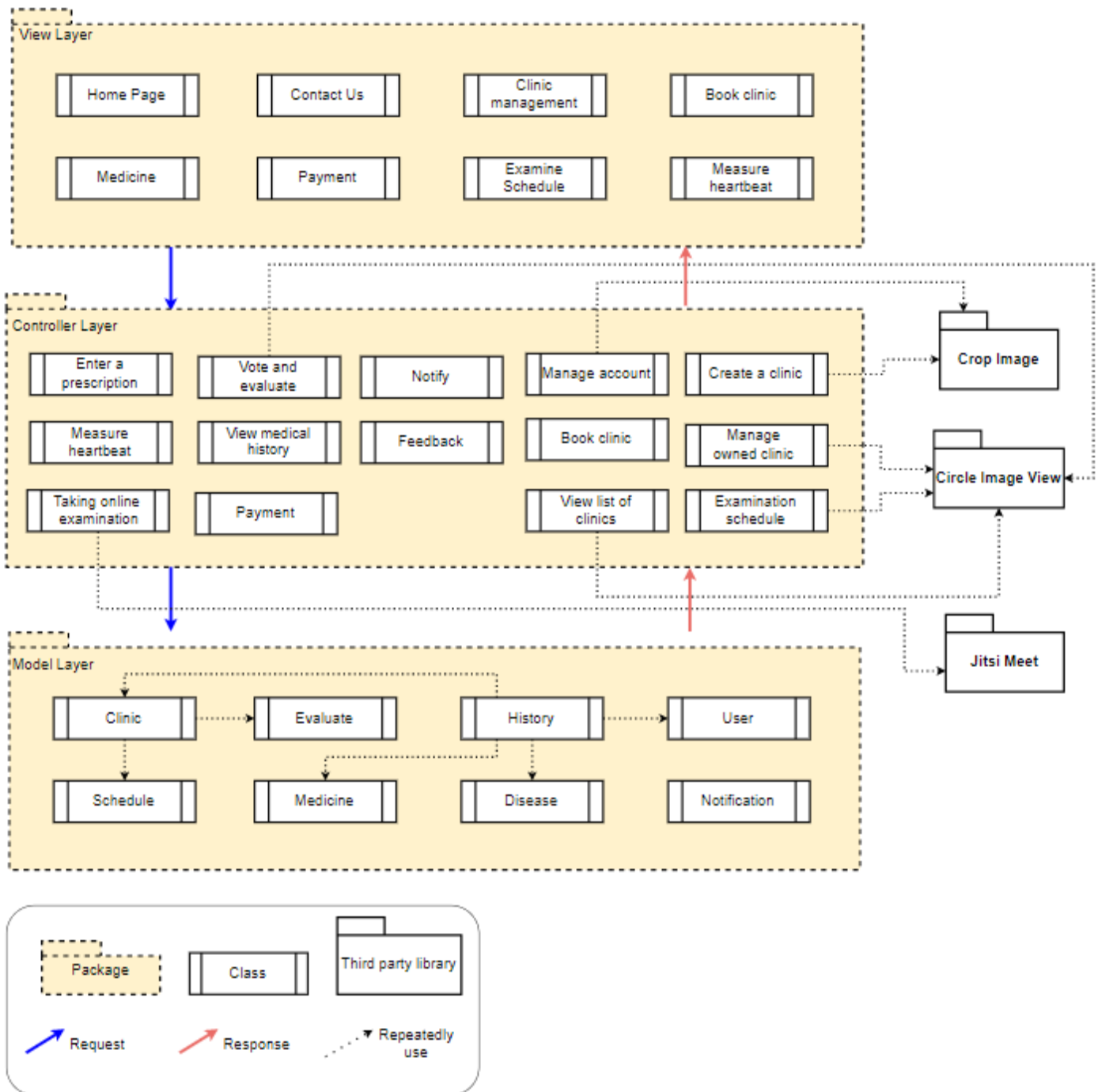


Figure 2: Module View

Explication

Element	Responsibilities
Model layer	Contains system service data
Controller layer	Controls to handle user actions
View layer	Where users interact directly
Crop image	Imported library to crop images
Circle image view	Imported library to round the image
Jitsi meet	A free and open-source video conferencing application

Description

The system consists of 3 layers: Model layer, Controller layer, and View layer.

Users will interact with system features (View layer), interactions will be handled through corresponding controls (Controller layer) to access corresponding data (Model layer).

3.3. Allocation view

The allocation view models the run-time architecture of a system. It shows the configuration of the hardware elements when the system is deployed.

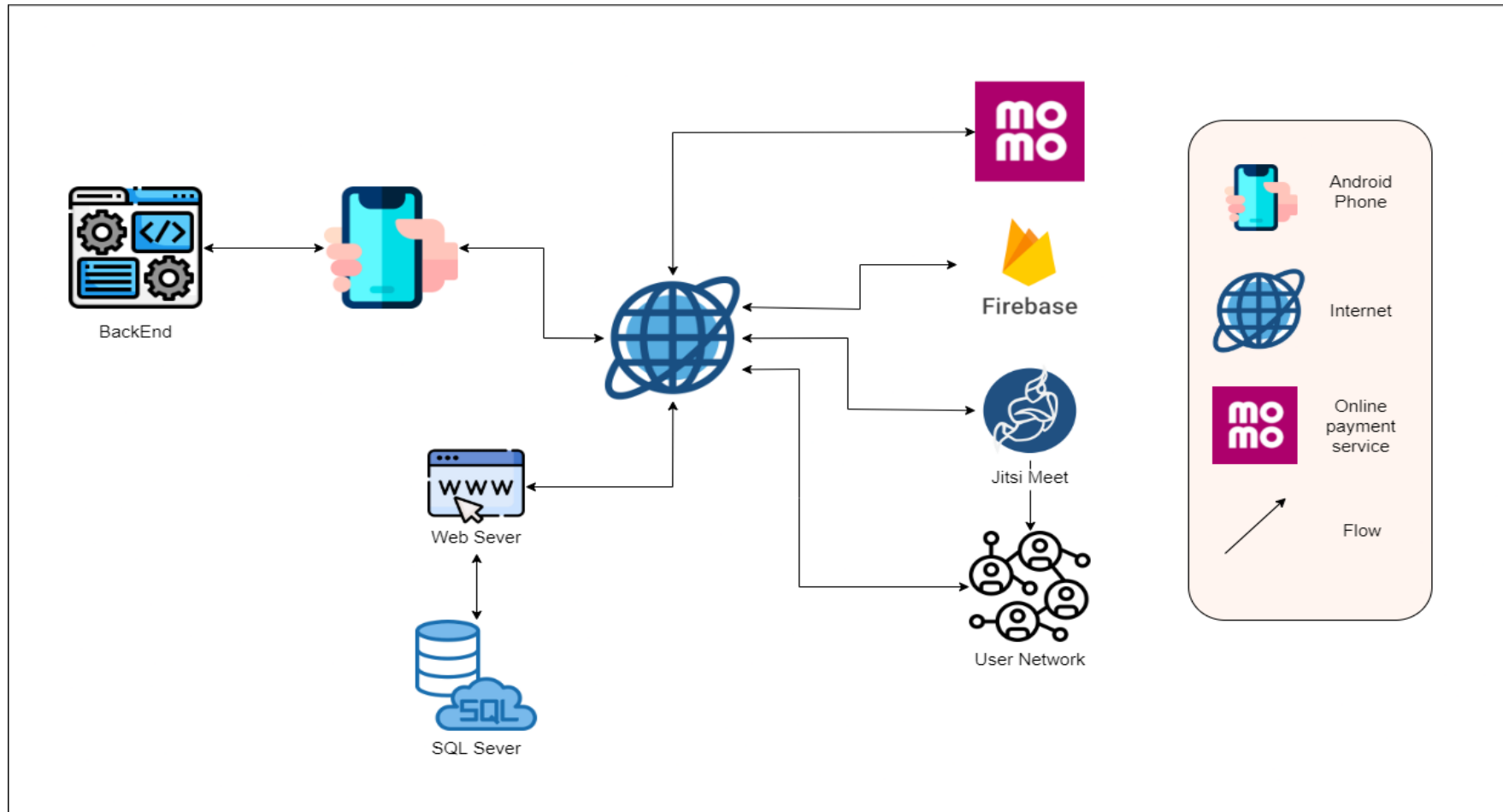


Figure 3: Allocation view

Explication

Element	Responsibilities
Android phone	The device is installed with the application, the user will interact with the system through this device
Back end	Contains codes to handle user actions
Internet	To connect the system to services
Firebase service	Service providing real-time data storage solutions
Jitsi meet	A free and open-source video conferencing application
User network	Network connecting users to use the application
Web service	Handle disease prediction function
SQL server	Contains data serving the webserver

Description

Users will interact with the system through an application on an internet-connected Android smartphone. Through the system, after being connected to the internet, users will use the system's services such as online payment, clinic lookup, online examination, disease prediction, personal information update...